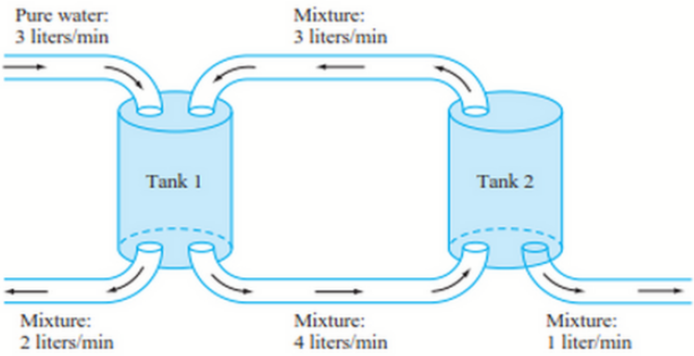
MA1513 Additional Practice Set 5

Problem 1

Two interconnected tanks are connected by pipes. Tank 1 initially contains 20 litres of water with 150g chlorine dissolved. Tank 2 initially contains 10 litres of water with 50g chlorine dissolved.



a) Derive the system of differential equations for $y_1(t)$ (chlorine mass in Tank 1) and $y_2(t)$ (chlorine mass in Tank 2).
b) Solve the system to find $y_1(t)$ and $y_2(t)$.
c) Determine the long-term chlorine concentration in each tank.

Problem 2

The general solution of an SDE $\boldsymbol{y}' = \boldsymbol{A}\boldsymbol{y}$ is given by

$$\boldsymbol{y} = c_1 e^{2t}\begin{pmatrix}1\\2\end{pmatrix} + c_2 e^{-3t}\begin{pmatrix}2\\-1\end{pmatrix}.$$

a) Find the matrix $\boldsymbol{A}$,
b) What is the type and stability of the equilibrium point?
c) Sketch the phase portrait of the SDE, indicating clearly the direction of each trajectory.

Problem 3

Given the second order ODE $4y'' + cy' + 8y = 0$ represents a damped mass-spring system.

a) Convert the ODE to a first-order SDE.
b) Determine the values of c for which the system exhibits under-damping.
c) Find the general solution of the ODE when $c = 8$.

Problem 4

Given the SDE $\boldsymbol{y}' = \boldsymbol{A}\boldsymbol{y}$ with $\boldsymbol{A} = \begin{pmatrix} 0 & 4 \\ -5 & -4 \end{pmatrix}$.

a) Construct a fundamental set of real solutions of the SDE.
b) Classify the type and stability of the equilibrium point.
c) Sketch a phase portrait of the system.

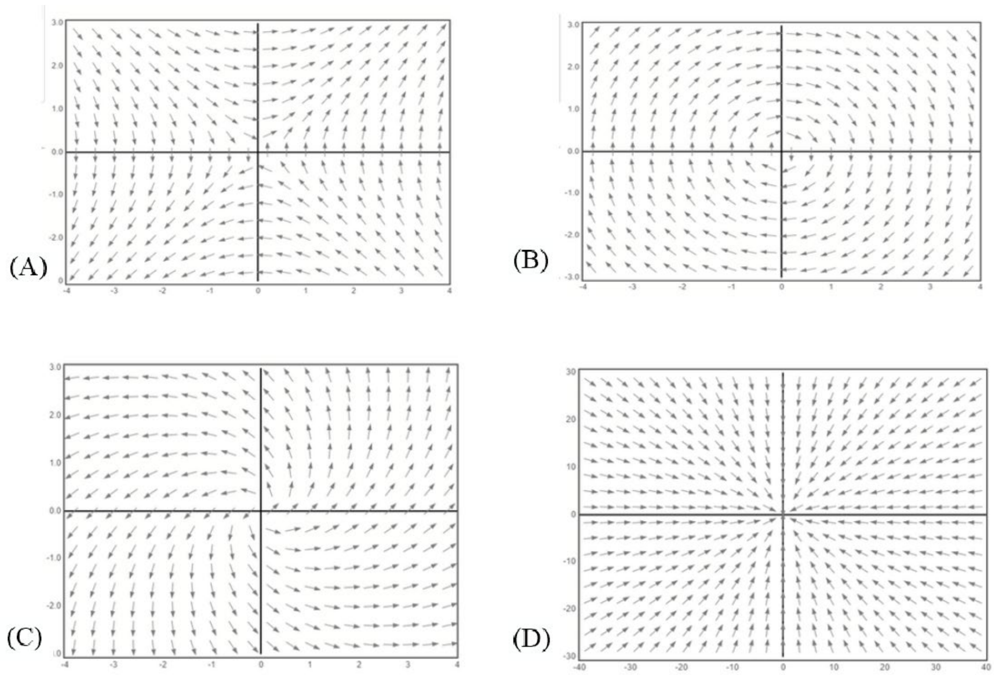
Problem 5

For the damped mass-spring system $my'' + cy' + ky = 0$ with $m = 1$ kg, and $k = 4$ N/m.

a) Convert the ODE to a first-order SDE in terms of c.
b) Find the value of c for which the system has a degenerate node.
c) Solve the system at this value of c in (b).
d) Suppose the initial position (t = 0) of the ball is y(0) = 0.5 m and the initial velocity is y'(0) = 0.5 m/s. Find the solution of the initial value problem for the ODE.

Problem 6

Given four direction fields (A–D):


(A) (B) (C) (D)

a) Identify the equilibrium type and stability of the equilibrium point corresponding to each direction field.
b) Which of the direction fields belong to the SDE $\begin{cases} y_1' &= y_2 \\ y_2' &= -y_1 \end{cases}$?

Problem 7

Given $\boldsymbol{A} = \begin{pmatrix} -2 & 0 \\ 3 & -2 \end{pmatrix}$, which has a single eigenvalue $\lambda = -2$ and eigenvector $\boldsymbol{v} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$:

a) Find a generalized eigenvector of $\boldsymbol{A}$ corresponding to $\boldsymbol{v}$.
b) Construct the general solution of $\boldsymbol{y}' = \boldsymbol{A}\boldsymbol{y}$.
c) Sketch the phase portrait of the SDE, indicating clearly the direction of each trajectory.
d) Solve the IVP given that $\boldsymbol{y}(0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Problem 8

Match the eigenvalue/eigenvector pairs (I) to (VI) with the correct direction fields A to F without using phase plane plotting app.

(I) $r_1 = -2,\ \ r_2 = 2;\ \mathrm{v}_1 = \begin{pmatrix} -1 \\ 3 \end{pmatrix},\ \mathrm{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

(II) $r_1 = 3,\ \ r_2 = 9;\ \mathrm{v}_1 = \begin{pmatrix} -3 \\ 1 \end{pmatrix},\ \mathrm{v}_2 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$

(III) $r_1 = 3\sqrt{2}i,\ \ r_2 = -3\sqrt{2}i;\ \mathrm{v}_1 = \begin{pmatrix} \sqrt{2} \\ 3i \end{pmatrix},\ \mathrm{v}_2 = \begin{pmatrix} \sqrt{2} \\ -3i \end{pmatrix}$

(IV) $r_1 = 0,\ \ r_2 = -2;\ \mathrm{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix},\ \mathrm{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

(V) $r_1 = 1,\ \ r_2 = 1;\ \mathrm{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

(VI) $r_1 = 2 + 2i,\ \ r_2 = 2 - 2i;\ \mathrm{v}_1 = \begin{pmatrix} 1 \\ -i \end{pmatrix},\ \mathrm{v}_2 = \begin{pmatrix} 1 \\ i \end{pmatrix}$